

Series and Integral Transforms

Theorems & Formulas Reference Sheet

Course 21183 — HIT, Prof. Michael Kroyter

Contents

1 Numerical Series

1.1 Basic Definitions

- The series $\sum_{n=1}^{\infty} a_n$ **converges** if the partial sums $S_N = \sum_{n=1}^N a_n$ have a finite limit.
- **Necessary condition:** if $\sum a_n$ converges, then $a_n \rightarrow 0$.
Contrapositive: if $a_n \not\rightarrow 0$, the series diverges.

1.2 Convergence Tests

Test	Statement
Comparison	If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, so does $\sum a_n$.
Limit Comparison	If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \in (0, \infty)$, both series behave the same.
Ratio (D'Alembert)	$L = \lim \left \frac{a_{n+1}}{a_n} \right $: converges if $L < 1$, diverges if $L > 1$, inconclusive if $L = 1$.
Root (Cauchy)	$L = \limsup a_n ^{1/n}$: converges if $L < 1$, diverges if $L > 1$.
Integral	$\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x) dx$ converge/diverge together (for f positive and decreasing).
Alternating (Leibniz)	$\sum (-1)^n b_n$ converges if $b_n \geq 0$ and $b_n \searrow 0$.
Dirichlet	$\sum a_n b_n$ converges if partial sums of a_n are bounded and $b_n \searrow 0$.
Abel	$\sum a_n b_n$ converges if $\sum a_n$ converges and b_n is monotone and bounded.

1.3 Absolute vs. Conditional Convergence

- **Absolutely convergent:** $\sum |a_n| < \infty$ (implies convergent).
- **Conditionally convergent:** $\sum a_n$ converges but $\sum |a_n|$ diverges.

1.4 p -Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges if and only if } p > 1.$$

2 Power Series

2.1 Radius of Convergence

The series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges absolutely for $|x-a| < R$, where

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}} \quad (\text{Cauchy-Hadamard formula})$$

Also: $R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ when the limit exists.

2.2 Taylor / Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

2.3 Common Maclaurin Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad R = \infty$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \quad |x| \leq 1, \quad x \neq -1$$

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad |x| < 1$$

3 Fourier Series

3.1 Definition

For f periodic with period $2L$:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Coefficients:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

3.2 Dirichlet Conditions

If f is piecewise smooth and periodic, the Fourier series converges to:

- $f(x)$ at points of continuity,
- $\frac{f(x^+) + f(x^-)}{2}$ at jump discontinuities.

3.3 Even and Odd Functions

- **Even** f : $b_n = 0$ (cosine series only).
- **Odd** f : $a_n = 0$ (sine series only).

3.4 Parseval's Identity

$$\frac{1}{L} \int_{-L}^L |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

3.5 Complex Form

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

4 Fourier Transform

4.1 Definition

$$\hat{f}(\xi) = \mathcal{F}\{f\}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx$$

(Alternative convention: $\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$)

4.2 Key Properties

Property	Formula
Linearity	$\mathcal{F}\{af + bg\} = a\hat{f} + b\hat{g}$
Shift	$\mathcal{F}\{f(x - a)\} = e^{-2\pi i a \xi} \hat{f}(\xi)$
Scaling	$\mathcal{F}\{f(ax)\} = \frac{1}{ a } \hat{f}(\xi/a)$
Derivative	$\mathcal{F}\{f'\}(\xi) = 2\pi i \xi \hat{f}(\xi)$
Convolution	$\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$

4.3 Parseval / Plancherel Theorem

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

5 Laplace Transform

5.1 Definition

$$\mathcal{L}\{f\}(s) = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

5.2 Common Transforms

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(bt)$	$\frac{b}{s^2 + b^2}$
$\cos(bt)$	$\frac{s}{s^2 + b^2}$
$e^{at} f(t)$	$F(s-a)$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$

5.3 Convolution Theorem

$$\mathcal{L}\{f * g\} = F(s) \cdot G(s)$$